EEE4114F Class Test 1

March 2026

Duration: 1 hour

Total: 20 marks

Empl ID: ______________________________

Instructions:

- This is a closed-book test.
- There are *four* questions totalling 20 marks. Answer all questions.
- Questions should be answered in the spaces provided on the test script. If necessary, you may also write on the backs of the printed pages, but please indicate where you have done so.
- You must use a pen to complete the exam – pencil will only be marked for drawings. Values indicated on drawings must be written in pen.
- Please show all of your working clearly – your method and approach matter.
- Do not write your student number or name anywhere on this script; instead, provide only the 7-digit EMPLID found on your student card.
- A formula sheet is provided with the paper.
- You have 1 hour.

Question	Marks	Grade
1	7	
2	5	
3	5	
4	3	
Total:	20	

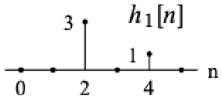
Question 1 [7 marks]

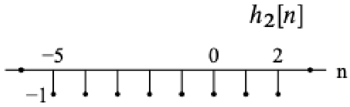
Consider the two impulse responses below:

$$h_1[n] = 3\delta[n-2] + \delta[n-4]$$
$$h_2[n] = u[n-3] - u[n+5]$$

1. Without using any transforms, determine which of the impulse responses describes a **causal** and **stable** LTI processor. Show all working. (4)

Solution: The impulse responses are sketched below:





Since $h_1[n] = 0$ for $n < 0$, it is the impulse response of a causal system. However, the system described by $h_2[n]$ is not causal since the impulse response is not zero for *all* $n < 0$.
Both impulse responses are absolutely summable:

$$S_1 = \sum_{n=-\infty}^{\infty} |h_1[n]| = 4 \quad \text{and} \quad S_2 = \sum_{n=-\infty}^{\infty} |h_2[n]| = 8.$$

And thus both systems are stable. As a result, only the first impulse response corresponds to a causal and stable LTI system.

2. Find an expression for the step response of the first system (with impulse response $h_1[n]$), and then sketch it. (3)

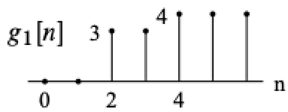
Solution: The step response is the convolution of the impulse response with the unit step, or equivalently, the accumulated impulse response:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]u[n-k] = \sum_{k=-\infty}^{n} h[k]u[n-k] = \sum_{k=-\infty}^{n} h[k],$$

Note: Students do not need to derive the above equivalence, but they must show some kind of logic in determining the final expression and sketch.
The step response for the first system is thus

$$g_1[n] = 3\delta[n-2] + 3\delta[n-3] + 4u[n-4]$$

This is sketched as follows:



Question 2 [5 marks]
Find the DTFT of the sequence below given that $|\alpha| < 1$.

$$y[n] = \alpha^{|n|}$$

Solution: The required transform is

$$\begin{aligned}
Y(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} \alpha^{|n|} e^{-j\omega n} \\
&= \sum_{n=-\infty}^{0} \alpha^{|n|} e^{-j\omega n} + \sum_{n=0}^{\infty} \alpha^{|n|} e^{-j\omega n} - \alpha^{|0|} e^{0} \\
&= \sum_{n=-\infty}^{0} \alpha^{-n} e^{-j\omega n} + \sum_{n=0}^{\infty} \alpha^{n} e^{-j\omega n} - 1 \\
&= \sum_{n=0}^{\infty} \alpha^{n} e^{j\omega n} + \sum_{n=0}^{\infty} \alpha^{n} e^{-j\omega n} - 1 \\
&= \sum_{n=0}^{\infty} (\alpha e^{j\omega})^{n} + \sum_{n=0}^{\infty} (\alpha e^{-j\omega})^{n} - 1 \\
&= \frac{1}{(1 - \alpha e^{j\omega})} + \frac{1}{(1 - \alpha e^{-j\omega})} - 1
\end{aligned}$$

since we know that each infinite sums exists for $|\alpha| < 1$, and $|e^{j\omega}| = |e^{-j\omega}| = 1$.

Question 3 [5 marks]
An LTI system is described by the difference equation

$$y[n] - 0.2y[n-1] = x[n] + 0.5x[n-1].$$

Determine the impulse response of the system under the assumption that it is causal.

Solution: The z-transform of the difference equation is

$$Y(z)(1 - 0.2z^{-1}) = X(z)(1 + 0.5z^{-1})$$

so

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + 0.5z^{-1}}{1 - 0.2z^{-1}} = \frac{z + 0.5}{z - 0.2}.$$

The system has a pole at $z = 0.2$, and thus the ROC for a causal system will be outside this pole, i.e., $|z| > 0.2$. We can thus find the inverse z-transform by writing

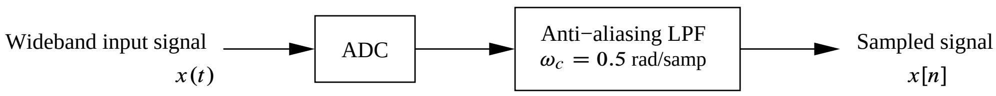
$$H(z) = \frac{1 + 0.5z^{-1}}{1 - 0.2z^{-1}} = \frac{1}{1 - 0.2z^{-1}} + \frac{0.5z^{-1}}{1 - 0.2z^{-1}}$$

and thus, by inspection,

$$h[n] = (0.2)^n u[n] + 0.5(0.2)^{(n-1)} u[n-1].$$

Question 4 [3 marks]

Consider the following sampling system:



Briefly explain what is wrong with this configuration. What consequences could this have?

Solution:

The anti-aliasing filter should be placed *before* the ADC, and must hence be an analog filter with a cut-off frequency in [rad/sec]. With the current configuration, aliasing may occur at the ADC, and thus the discrete-time "anti-aliasing" filter will not have the desired effect.